skaters: online distributional time-series forecasting by conjugation, in Python and JavaScript

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Summary

`skaters` is an online, univariate time-series forecaster whose every prediction is a full probability distribution. It is built by *conjugation*: an invertible transform wraps an inner forecaster, pushing each observation forward, forecasting in the transformed space, and pulling the predictive distribution back through the inverse. Because the inner forecaster can itself be a conjugation, models nest onto a single distributional leaf, and an ensemble is just another forecaster.

The library exposes one factory, `laplace`, with no required tuning parameters. Each call consumes one observation and returns k predictive distributions plus a calibration state: the probability integral transform and normalized surprise of the arriving point under the forecasts previously issued for it. Since v0.13.0 every predictive carries generalized-Pareto tails fitted online by censored maximum likelihood [@diks2011likelihood], so stated tail probabilities match measured frequencies: on 142 economic series the nominal 99.9% central interval covers 99.85% empirically, where a Gaussian read covers 99.13%.

The package is written twice, in pure Python with no dependencies (`pip install skaters`) and in zero-dependency JavaScript (`npm install skaters`). A parity suite checks about 100,000 probe values to 10^{-6} agreement on every run, so a model trained on a server continues bit-compatibly in a browser via Pyodide [@pyodide2021].

Statement of need

Density forecasts are reusable objects: the same predictive supports pricing, risk limits, threshold alarms, and simulation. Most accessible forecasting software returns point forecasts or intervals, refits in batch, or requires a scientific-computing stack. Practitioners who need a predictive *distribution* per tick, updated online at microsecond-to-second cadence, on a server or in a browser, have had to assemble one from parts. `skaters` supplies one as a single dependency-free function. On 894 continuous non-price FRED series it leads classical, neural, and foundation-model baselines on held-out log-likelihood under a rolling one-step protocol; on asset prices a GARCH-t model remains preferable, a split reported rather than averaged away. The calibration state turns the forecaster into an anomaly detector with stated false-alarm rates: alarm when $\mathrm{erfc}(|z|/\sqrt{2}) < \alpha$, with measured rates near nominal on economic series. Methods, benchmarks, and the tail-calibration studies are documented in a companion methods paper in the repository and at https://skaters.microprediction.org.

State of the field

Established open-source forecasters occupy different corners: `statsmodels` [@seabold2010statsmodels] and `darts` [@herzen2022darts] are batch-refit and largely point- or interval-oriented, Prophet [@taylor2018forecasting] targets batch seasonal decomposition, `GluonTS` [@alexandrov2020gluonts] is distributional but assumes a deep-learning stack and offline training, and `river` [@montiel2021river] is natively online but centred on point prediction. None provides an online predictive density as a single dependency-free function, and none runs unchanged in a browser. The design distills ideas from the author's earlier `timemachines` package [@cotton2021timemachines] and from live distributional-prediction contests on the microprediction platform [@cotton2022microprediction], where multi-level prediction of residual streams anticipated the conjugation recursion. The repository's benchmark harness compares against representatives of each corner (AutoARIMA, AutoETS, the reference R forecasters, GARCH-t, neural and zero-shot foundation baselines) under a common rolling protocol, and the split of regimes is reported rather than averaged away.

Software design

Every forecaster satisfies one protocol: a pure function `(y, state) -> (dists, state)` that consumes one observation and returns k predictive distributions plus its own next state. State is an explicit dictionary, never hidden in an object, so a model can be checkpointed as JSON, resumed elsewhere, or handed across a language boundary mid-stream. Conjugation is function composition under this protocol: a transform wraps an inner skater, and because the wrapped object satisfies the same protocol, models nest to any depth and an ensemble is just another skater. The public surface is one factory, `laplace`, whose defaults require no tuning. The package has no runtime dependencies in any of its five ports (Python, JavaScript, R, Julia, Rust); the ports are locked to the Python reference by a parity suite of roughly 100,000 probe values at 10^{-6} agreement, run in continuous integration, which is what allows a model trained on a server to continue bit-compatibly in a browser via Pyodide [@pyodide2021].

Research impact statement

The package is the measurement instrument in an ongoing empirical study of representation value in conformal prediction, whose experiment scripts live in this repository's `benchmarks` directory, and the companion methods paper reports the underlying benchmark across 5,402 FRED series. Beyond its own results, the harness gives researchers a reproducible rolling-protocol comparison of distributional forecasters under log-likelihood and CRPS, with per-regime splits, and the calibration state gives applied users anomaly alarms with stated false-alarm rates. The browser build is used for interactive teaching demonstrations at https://skaters.microprediction.org.

AI usage disclosure

Portions of the source code, the multi-language ports, the parity and test suites, and drafts of this manuscript were produced with the assistance of a large language model (Claude, Anthropic) operated interactively by the author, who directed the design, reviewed the output, and takes responsibility for the content. The numerical claims are reproducible from the scripts in `benchmarks/`, and the cross-language parity suite checks the ports against the Python reference on every continuous-integration run.

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References